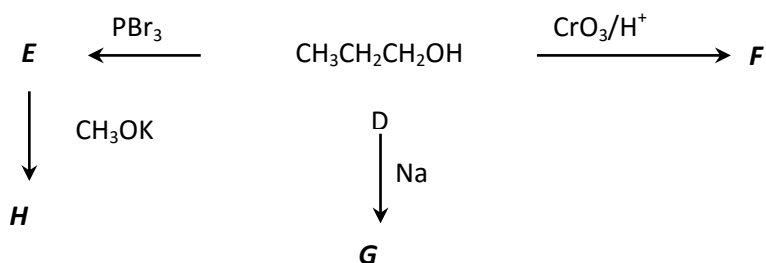


## Chapter 8: Hydroxy Compounds

PSPM 2011 / 2012

Q. 1 Reaction of 1-propanol, **D**, with several reagents is as shown below:

15m

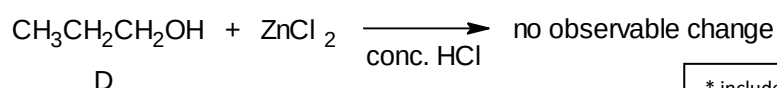
- (a) Draw the structures of **E**, **F**, **G** and **H**.  
 (b) Suggest another reagent to convert **D** to **F**.  
 (c) Give one chemical test to differentiate compound **D** from 2-methyl-2-butanol. State the observation and write the chemical equation.

Answer (a)  $\text{E} = \text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$   
 $\text{F} = \text{CH}_3\text{CH}_2\text{COOH}$   
 $\text{G} = \text{CH}_3\text{CH}_2\text{CH}_2\text{ONa}$   
 $\text{H} = \text{CH}_3\text{CH}_2\text{CH}_2\text{OCH}_3$

(b)  $\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}^+$  or  $\text{KMnO}_4/\text{H}^+$

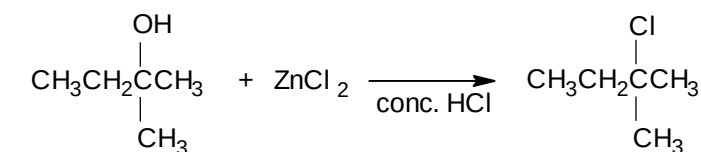
(c) Lucas Test or  $\text{ZnCl}_2$ , concentrated  $\text{HCl}$  @

Oxidation Test,  $\frac{\text{KMnO}_4/\text{H}^+}{\Delta}$  or  $\frac{\text{Na}_2\text{Cr}_2\text{O}_7/\text{H}^+}{\Delta}$



Observation: No observable change / forms one layer

\* include equation &  
observation



2-methyl-2-butanol

Observation: Cloudy solution form immediately / gives two layers immediately

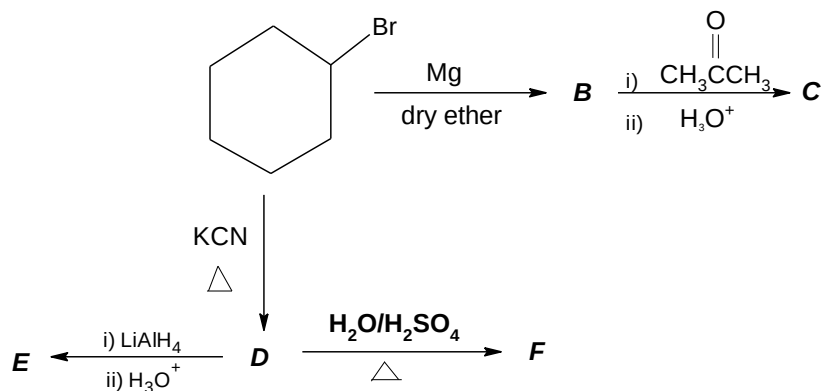
## PSPM 2012 / 2013

Q. 2 (a) Give reason for each of the following statements: 5 m

- (i) 1-Butanol is less acidic than phenol.
- (ii)  $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$  has higher boiling point than  $\text{CH}_3\text{COCOCH}_3$ .
- (iii) 2-methyl-2-propanol cannot undergo oxidation.
- (iv) 2-hexanol gives yellow precipitate with iodine in NaOH solution.
- (v) Both phenol and 1-propanol react with sodium but only phenol dissolves in aqueous NaOH.

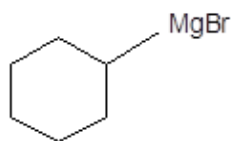
- Answer (a) (i) **Phenoxide ion** can be stabilized by **delocalization of electron @ resonance effect** into benzene ring, easier to be formed than butoxide ion.
- (ii)  $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$  has -OH group which can form **more hydrogen** bonds between molecules
- (iii) Because it is  $3^\circ$  alcohol
- (iv) 2-Hexanol has **methyl hydroxyl** group / methyl carbinol group which gives positive results with iodoform test
- (v) Phenol is more acidic than 1-propanol **OR** 1-propanol is less acidic

Q. 2 (b) Draw the structure of the products **B** through **F** in the reaction below. 5 m

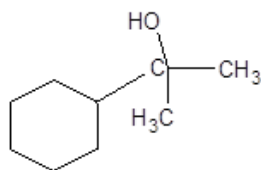


Answer (b)

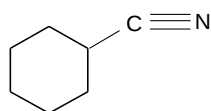
B =



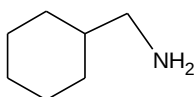
C =



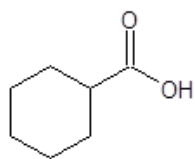
D =



E =

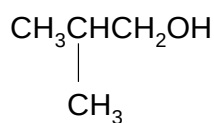


F =

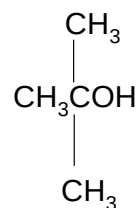


Q. 2 (c) Given the following alcohol:

5 m



Alcohol **G**



Alcohol **H**

- Name the chemical test to distinguish between the alcohol above.
- Describe your observation.
- Write the chemical equation for the observed reaction.
- Show how would you prepare alcohol **G** *via* a Grignard reagent.

Answer (c) (i) Lucas test @ Oxidation Test

(ii) LucasTest

Alcohol G : no observable change **within** 10 minutes

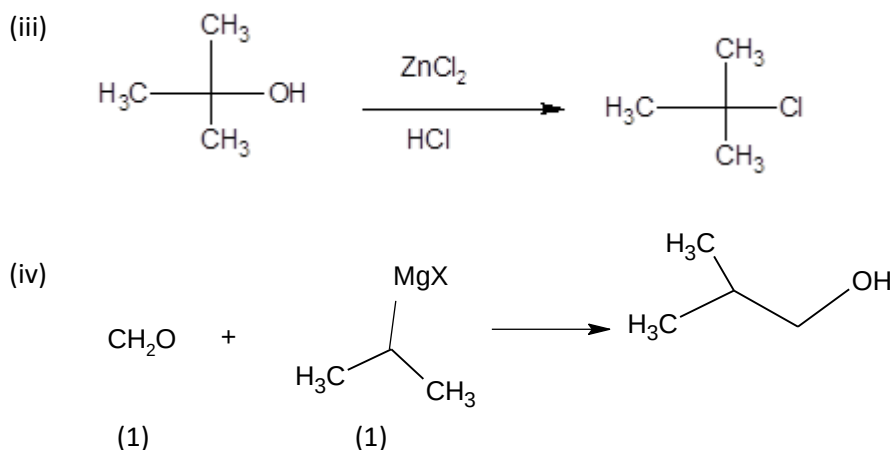
Alcohol H : cloudy solution formed immediately.

@

Oxidation Test

Alcohol G : purple to colourless

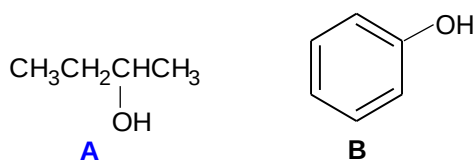
Alcohol H : no observable change



## PSPM 2013 / 2014

Q. 3 (a) The structure of compound **A** and **B** are as shown below.

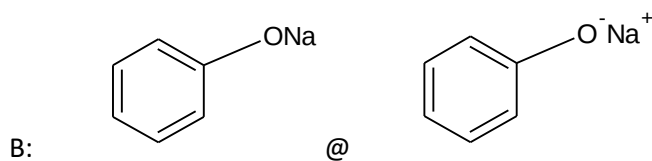
7 m



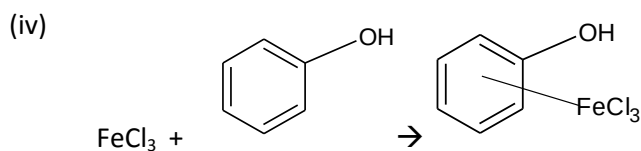
- Name the class of compound **A** and **B**.
- Draw the structure of product formed when **A** and **B** react with sodium hydroxide solution respectively.
- Compare the acidity of **A** and **B**. Explain your answer.
- Write the chemical equation when **B** reacts with  $\text{FeCl}_3$  solution.

Answer (a)(i) A: alcohol  
B: phenol @ aromatic alcohol

(ii) A: no reaction @ no product @ X

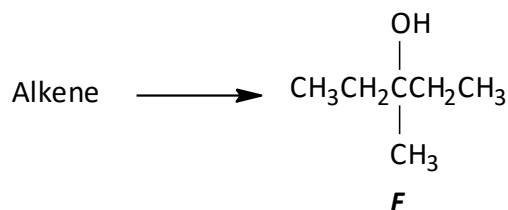


(iii) Phenol is **more acidic** than alcohol  
phenol can **delocalise its electrons** @  
phenoxide ion is **more stable** than alkoxide ion



## PSPM 2014 / 2015

- Q. 4 (a) Consider the following reaction scheme.

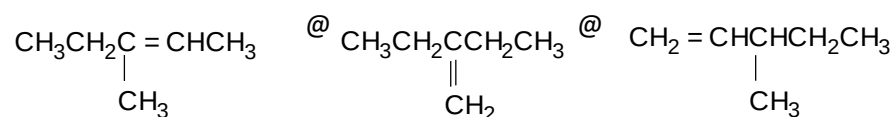


7 m

- (i) Give the IUPAC name for **F**.
- (ii) Suggest one possible alkene to produce **F** and give the suitable reagent for this conversion.
- (iii) Name a chemical test to differentiate between 1-pentanol, 2-pentanol and **F**. State all the observations.

Answer (b) (i) **F**: 3-methyl-3-pentanol @ 3-methylpentan-3-ol

- (ii) Alkene:



Reagent:  $\text{H}_2\text{O}, \text{H}^+$  @  $\text{H}_2\text{O}, \text{H}_2\text{SO}_4$  @  $\text{H}_2\text{O}, \text{H}_3\text{O}^+$

- (iii) Lucas test

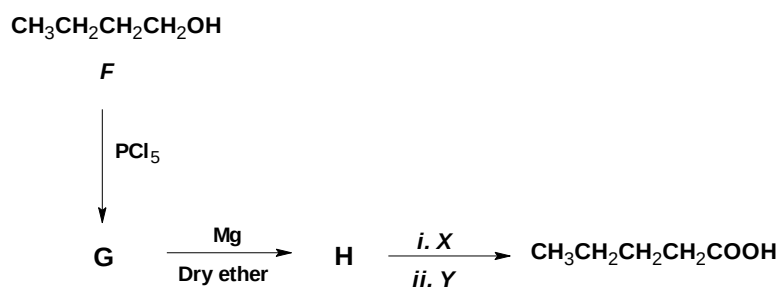
1-pentanol – no observable change

2-pentanol – cloudy solution formed within 5 minutes

**F** – cloudy solution form immediately

## PSPM 2015 / 2016

Q. 5 (a) Pentanoic acid can be prepared from 1-butanol, F, as shown below:



7 m

- (i) Draw the structures of G and H.
- (ii) State reagents X and Y.
- (iii) Give the reagent needed to prepare F from 1-butene.
- (iv) Suggest an alternative two-step reaction to convert G to pentanoic acid.

Answer (c) (i) G:  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Cl}$

H:  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{MgCl}$

(ii) X:  $\text{CO}_2$

Y:  $\text{H}_3\text{O}^+ @ \text{H}_2\text{O} / \text{H}^+ \text{H}_2\text{O}, \text{H}^+$

(iii) (i)  $\text{HBr}$ ,  $\text{H}_2\text{O}_2$ , (ii)  $\text{NaOH}$

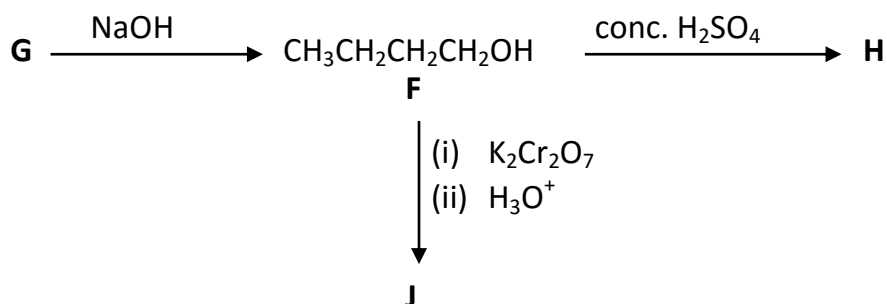
(iv)

$$\text{G} \xrightarrow[\text{@ KCN}]{\text{CN}^-} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CN} \xrightarrow[\text{@ NaCN}]{\text{H}_3\text{O}^+} \text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{COOH}$$

PSPM 2016 / 2017

Q. 6 (a) The reaction of 1-butanol, **F**, with several reagents is shown below.

11 m



- Draw the structures of **G**, **H** and **J**.
- Suggest the reagent(s) needed to prepare **F** from methanal.
- Name the functional group in compound **H**.
- Give one chemical test to differentiate compound **F** from 2-methyl-2-butanol. State the observation and write the chemical equation.
- Suggest the reagent(s) to convert **F** to butanal.

Answer (a) (i) **G** :  $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{X}$  (X=Cl @ Br @ I )  
**H** :  $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$  @  $\text{CH}_3\text{CH}=\text{CHCH}_3$   
**J** :  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$

(ii) (i)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{MgX}$ , (X=Cl @ Br ), (ii)  $\text{H}_2\text{O}/\text{H}^+$  @  $\text{H}_3\text{O}^+$   
 Carbon-carbon double bond

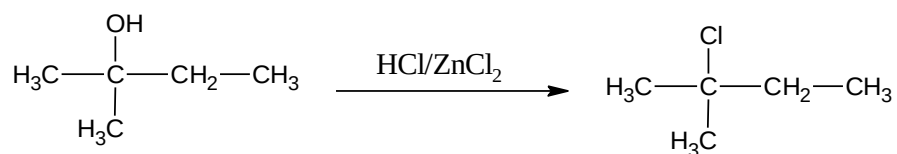
(iii) **Lucas Test**

(iv) Observation :

Compound **F** – No cloudy solution @ No observable changes @ Cloudy solution forms after 10 minutes

2-methyl-2-butanol – Immediate appearance of cloudy solution/emulsion

Chemical equation :

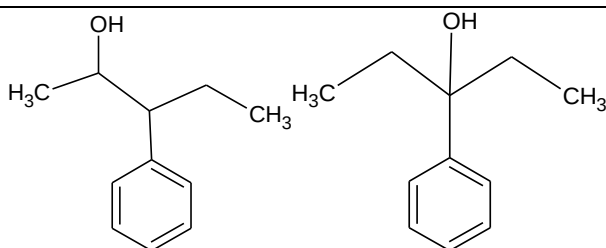


(v) PCC (pyridiniumchlorocromate)

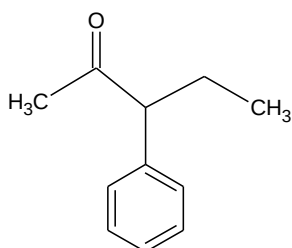
## PSPM 2017 / 2018

- Q. 7 (a) 3-phenylpentene reacts with acidified water to yield two compounds, H and J. the formation of compound H follows Markovnikov's rules whereas carbocation rearrangement forms compound J. 5 m
- (i) Draw the structural formulae for compounds H and J.
- (ii) Suggest the product formed when compound H is oxidized.
- (iii) What will happen when compound J is oxidized with acidified  $\text{KMnO}_4$  solution? Explain your answer.

Answer (a)(i)



- (ii) 3-phenyl-2-pentanone



- (iii)

no changes/ no reaction/ purple colour remain unchanged.

Because there is no benzylic hydrogen/ tertiary alcohol/no hydrogen attach to carbinol carbon.

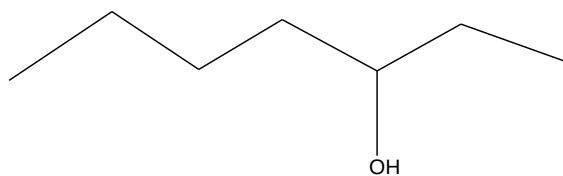


PSPM 2018 / 2019

Q. 8

The structural formula of compound H is shown below.

7 m



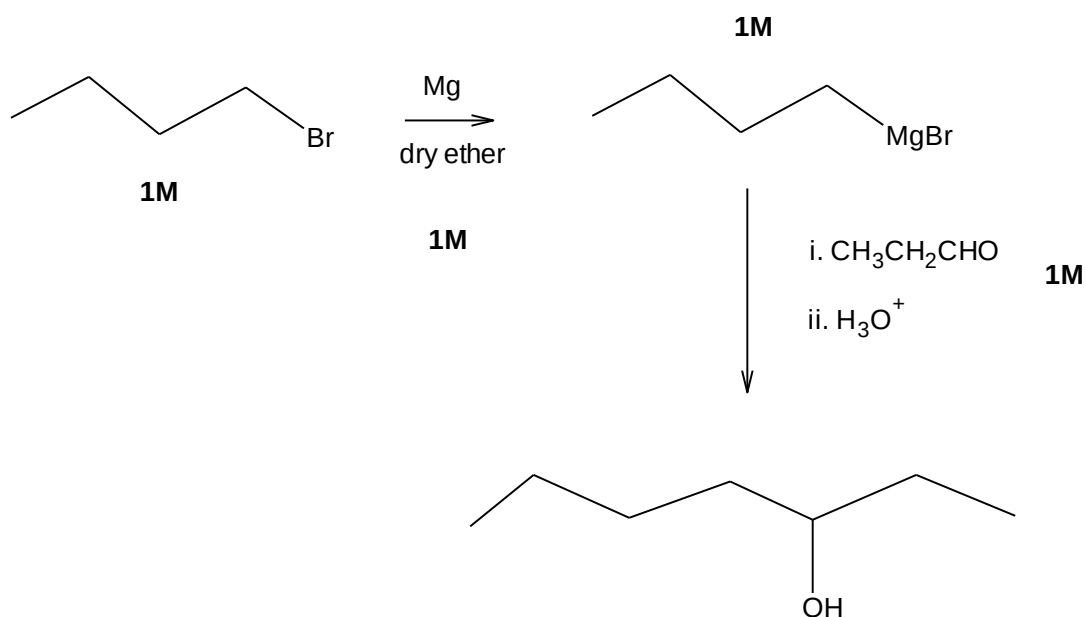
Compound H

- (a) Name the functional group of compound H.
- (b) Suggest a synthetic pathway to prepare H via Grignard reagent and propanal.
- (c) Compound H reacts with J to form K,  $C_{10}H_{20}O_2$ , in the presence of a few drops of sulphuric acid. Draw the structural formula of J. state the function of sulphuric acid.

Answer

- (a) hydroxyl

(b)



- (c) J :  $\text{CH}_3\text{CH}_2\text{COOH}$

Catalyst – Speed up the reaction